Forecast daily bike rental demand using time series models

Isha Gupta

2026-03-17

About Data Analysis Report

This RMarkdown file contains the report of the data analysis done for the project on forecasting daily bike rental demand using time series models in R. It includes data exploration, summary statistics, time series visualization, smoothing techniques, decomposition, stationarity testing, and forecasting using ARIMA models.

The final report was completed on Tue Mar 17 17:39:53 2026.

Data Description:

This dataset contains the daily count of rental bike transactions between 2011 and 2012 in the Capital Bikeshare system along with weather and seasonal information.

Data Source:
https://archive.ics.uci.edu/ml/datasets/bike+sharing+dataset
(https://archive.ics.uci.edu/ml/datasets/bike+sharing+dataset)

Relevant Paper:
Fanaee-T, Hadi, and Gama, Joao. Event labeling combining ensemble detectors and background knowledge, Progress in Artificial Intelligence (2013).

Task One: Load and Explore the Data

Load data and install packages

```
library(timetk)
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.2     ✔ readr     2.1.4
## ✔ forcats   1.0.0     ✔ stringr   1.5.0
## ✔ ggplot2   3.4.2     ✔ tibble    3.2.1
## ✔ lubridate 1.9.2     ✔ tidyr     1.3.0
## ✔ purrr     1.0.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
library(lubridate)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)
library(TTR)
```

Load Dataset

```
data("bike_sharing_daily")
bike_data <- bike_sharing_daily
head(bike_data)
```

```
## # A tibble: 6 × 16
##   instant dteday     season    yr  mnth holiday weekday workingday weathersit
##     <dbl> <date>      <dbl> <dbl> <dbl>   <dbl>   <dbl>      <dbl>      <dbl>
## 1       1 2011-01-01      1     0     1       0       6          0          2
## 2       2 2011-01-02      1     0     1       0       0          0          2
## 3       3 2011-01-03      1     0     1       0       1          1          1
## 4       4 2011-01-04      1     0     1       0       2          1          1
## 5       5 2011-01-05      1     0     1       0       3          1          1
## 6       6 2011-01-06      1     0     1       0       4          1          1
## # ℹ 7 more variables: temp <dbl>, atemp <dbl>, hum <dbl>, windspeed <dbl>,
## #   casual <dbl>, registered <dbl>, cnt <dbl>
```

```
summary(bike_data)
```

```
##     instant          dteday               season           yr        
##  Min.   :  1.0   Min.   :2011-01-01   Min.   :1.000   Min.   :0.0000  
##  1st Qu.:183.5   1st Qu.:2011-07-02   1st Qu.:2.000   1st Qu.:0.0000  
##  Median :366.0   Median :2012-01-01   Median :3.000   Median :1.0000  
##  Mean   :366.0   Mean   :2012-01-01   Mean   :2.497   Mean   :0.5007  
##  3rd Qu.:548.5   3rd Qu.:2012-07-01   3rd Qu.:3.000   3rd Qu.:1.0000  
##  Max.   :731.0   Max.   :2012-12-31   Max.   :4.000   Max.   :1.0000  
##       mnth          holiday           weekday        workingday   
##  Min.   : 1.00   Min.   :0.00000   Min.   :0.000   Min.   :0.000  
##  1st Qu.: 4.00   1st Qu.:0.00000   1st Qu.:1.000   1st Qu.:0.000  
##  Median : 7.00   Median :0.00000   Median :3.000   Median :1.000  
##  Mean   : 6.52   Mean   :0.02873   Mean   :2.997   Mean   :0.684  
##  3rd Qu.:10.00   3rd Qu.:0.00000   3rd Qu.:5.000   3rd Qu.:1.000  
##  Max.   :12.00   Max.   :1.00000   Max.   :6.000   Max.   :1.000  
##    weathersit         temp              atemp              hum        
##  Min.   :1.000   Min.   :0.05913   Min.   :0.07907   Min.   :0.0000  
##  1st Qu.:1.000   1st Qu.:0.33708   1st Qu.:0.33784   1st Qu.:0.5200  
##  Median :1.000   Median :0.49833   Median :0.48673   Median :0.6267  
##  Mean   :1.395   Mean   :0.49538   Mean   :0.47435   Mean   :0.6279  
##  3rd Qu.:2.000   3rd Qu.:0.65542   3rd Qu.:0.60860   3rd Qu.:0.7302  
##  Max.   :3.000   Max.   :0.86167   Max.   :0.84090   Max.   :0.9725  
##    windspeed           casual         registered        cnt      
##  Min.   :0.02239   Min.   :   2.0   Min.   :  20   Min.   :  22  
##  1st Qu.:0.13495   1st Qu.: 315.5   1st Qu.:2497   1st Qu.:3152  
##  Median :0.18097   Median : 713.0   Median :3662   Median :4548  
##  Mean   :0.19049   Mean   : 848.2   Mean   :3656   Mean   :4504  
##  3rd Qu.:0.23321   3rd Qu.:1096.0   3rd Qu.:4776   3rd Qu.:5956  
##  Max.   :0.50746   Max.   :3410.0   Max.   :6946   Max.   :8714  
```

```
names(bike_data)
```

```
##  [1] "instant"    "dteday"     "season"     "yr"         "mnth"      
##  [6] "holiday"    "weekday"    "workingday" "weathersit" "temp"      
## [11] "atemp"      "hum"        "windspeed"  "casual"     "registered"
## [16] "cnt"
```

```
data("bike_sharing_daily")

bike_data <- bike_sharing_daily

head(bike_data)
```

```
## # A tibble: 6 × 16
##   instant dteday     season    yr  mnth holiday weekday workingday weathersit
##     <dbl> <date>      <dbl> <dbl> <dbl>   <dbl>   <dbl>      <dbl>      <dbl>
## 1       1 2011-01-01      1     0     1       0       6          0          2
## 2       2 2011-01-02      1     0     1       0       0          0          2
## 3       3 2011-01-03      1     0     1       0       1          1          1
## 4       4 2011-01-04      1     0     1       0       2          1          1
## 5       5 2011-01-05      1     0     1       0       3          1          1
## 6       6 2011-01-06      1     0     1       0       4          1          1
## # ℹ 7 more variables: temp <dbl>, atemp <dbl>, hum <dbl>, windspeed <dbl>,
## #   casual <dbl>, registered <dbl>, cnt <dbl>
```

```
summary(bike_data)
```

```
##     instant          dteday               season            yr        
##  Min.   :  1.0   Min.   :2011-01-01   Min.   :1.000   Min.   :0.0000  
##  1st Qu.:183.5   1st Qu.:2011-07-02   1st Qu.:2.000   1st Qu.:0.0000  
##  Median :366.0   Median :2012-01-01   Median :3.000   Median :1.0000  
##  Mean   :366.0   Mean   :2012-01-01   Mean   :2.497   Mean   :0.5007  
##  3rd Qu.:548.5   3rd Qu.:2012-07-01   3rd Qu.:3.000   3rd Qu.:1.0000  
##  Max.   :731.0   Max.   :2012-12-31   Max.   :4.000   Max.   :1.0000  
##       mnth          holiday           weekday        workingday   
##  Min.   : 1.00   Min.   :0.00000   Min.   :0.000   Min.   :0.000  
##  1st Qu.: 4.00   1st Qu.:0.00000   1st Qu.:1.000   1st Qu.:0.000  
##  Median : 7.00   Median :0.00000   Median :3.000   Median :1.000  
##  Mean   : 6.52   Mean   :0.02873   Mean   :2.997   Mean   :0.684  
##  3rd Qu.:10.00   3rd Qu.:0.00000   3rd Qu.:5.000   3rd Qu.:1.000  
##  Max.   :12.00   Max.   :1.00000   Max.   :6.000   Max.   :1.000  
##    weathersit         temp              atemp              hum        
##  Min.   :1.000   Min.   :0.05913   Min.   :0.07907   Min.   :0.0000  
##  1st Qu.:1.000   1st Qu.:0.33708   1st Qu.:0.33784   1st Qu.:0.5200  
##  Median :1.000   Median :0.49833   Median :0.48673   Median :0.6267  
##  Mean   :1.395   Mean   :0.49538   Mean   :0.47435   Mean   :0.6279  
##  3rd Qu.:2.000   3rd Qu.:0.65542   3rd Qu.:0.60860   3rd Qu.:0.7302  
##  Max.   :3.000   Max.   :0.86167   Max.   :0.84090   Max.   :0.9725  
##    windspeed           casual         registered        cnt      
##  Min.   :0.02239   Min.   :   2.0   Min.   :  20   Min.   :  22  
##  1st Qu.:0.13495   1st Qu.: 315.5   1st Qu.:2497   1st Qu.:3152  
##  Median :0.18097   Median : 713.0   Median :3662   Median :4548  
##  Mean   :0.19049   Mean   : 848.2   Mean   :3656   Mean   :4504  
##  3rd Qu.:0.23321   3rd Qu.:1096.0   3rd Qu.:4776   3rd Qu.:5956  
##  Max.   :0.50746   Max.   :3410.0   Max.   :6946   Max.   :8714
```

```
str(bike_data)
```

```
## spc_tbl_ [731 × 16] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ instant   : num [1:731] 1 2 3 4 5 6 7 8 9 10 ...
##  $ dteday    : Date[1:731], format: "2011-01-01" "2011-01-02" ...
##  $ season    : num [1:731] 1 1 1 1 1 1 1 1 1 1 ...
##  $ yr        : num [1:731] 0 0 0 0 0 0 0 0 0 0 ...
##  $ mnth      : num [1:731] 1 1 1 1 1 1 1 1 1 1 ...
##  $ holiday   : num [1:731] 0 0 0 0 0 0 0 0 0 0 ...
##  $ weekday   : num [1:731] 6 0 1 2 3 4 5 6 0 1 ...
##  $ workingday: num [1:731] 0 0 1 1 1 1 1 0 0 1 ...
##  $ weathersit: num [1:731] 2 2 1 1 1 1 2 2 1 1 ...
##  $ temp      : num [1:731] 0.344 0.363 0.196 0.2 0.227 ...
##  $ atemp     : num [1:731] 0.364 0.354 0.189 0.212 0.229 ...
##  $ hum       : num [1:731] 0.806 0.696 0.437 0.59 0.437 ...
##  $ windspeed : num [1:731] 0.16 0.249 0.248 0.16 0.187 ...
##  $ casual    : num [1:731] 331 131 120 108 82 88 148 68 54 41 ...
##  $ registered: num [1:731] 654 670 1229 1454 1518 ...
##  $ cnt       : num [1:731] 985 801 1349 1562 1600 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   instant = col_double(),
##   ..   dteday = col_date(format = ""),
##   ..   season = col_double(),
##   ..   yr = col_double(),
##   ..   mnth = col_double(),
##   ..   holiday = col_double(),
##   ..   weekday = col_double(),
##   ..   workingday = col_double(),
##   ..   weathersit = col_double(),
##   ..   temp = col_double(),
##   ..   atemp = col_double(),
##   ..   hum = col_double(),
##   ..   windspeed = col_double(),
##   ..   casual = col_double(),
##   ..   registered = col_double(),
##   ..   cnt = col_double()
##   .. )
```

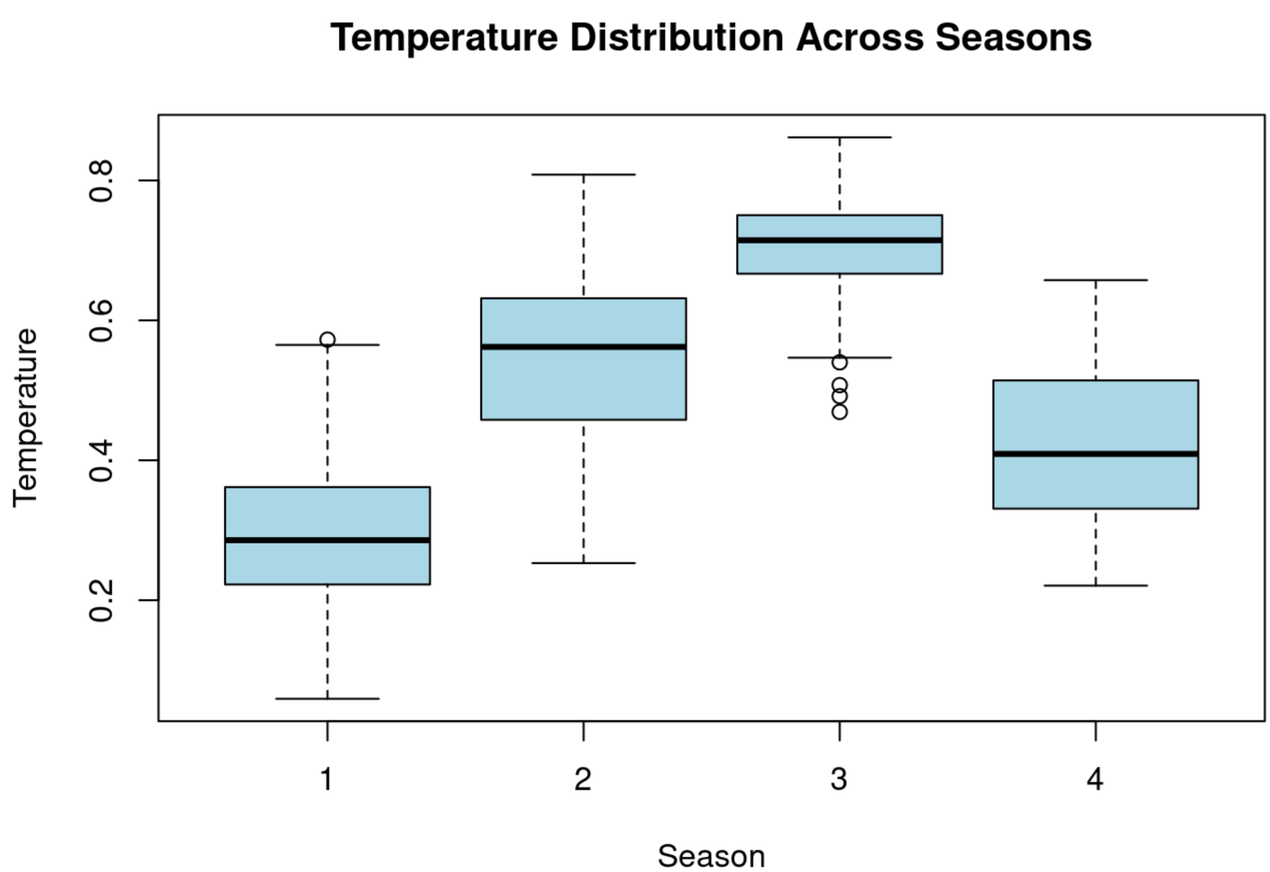
```
cor(bike_data$temp, bike_data$cnt)
```

```
## [1] 0.627494
```

```
cor(bike_data$atemp, bike_data$cnt)
```

```
## [1] 0.6310657
```

```
boxplot(temp ~ season,
        data = bike_data,
        col="lightblue",
        main="Temperature Distribution Across Seasons",
        xlab="Season",
        ylab="Temperature")
```




```
library(tidyverse)
library(lubridate)
```

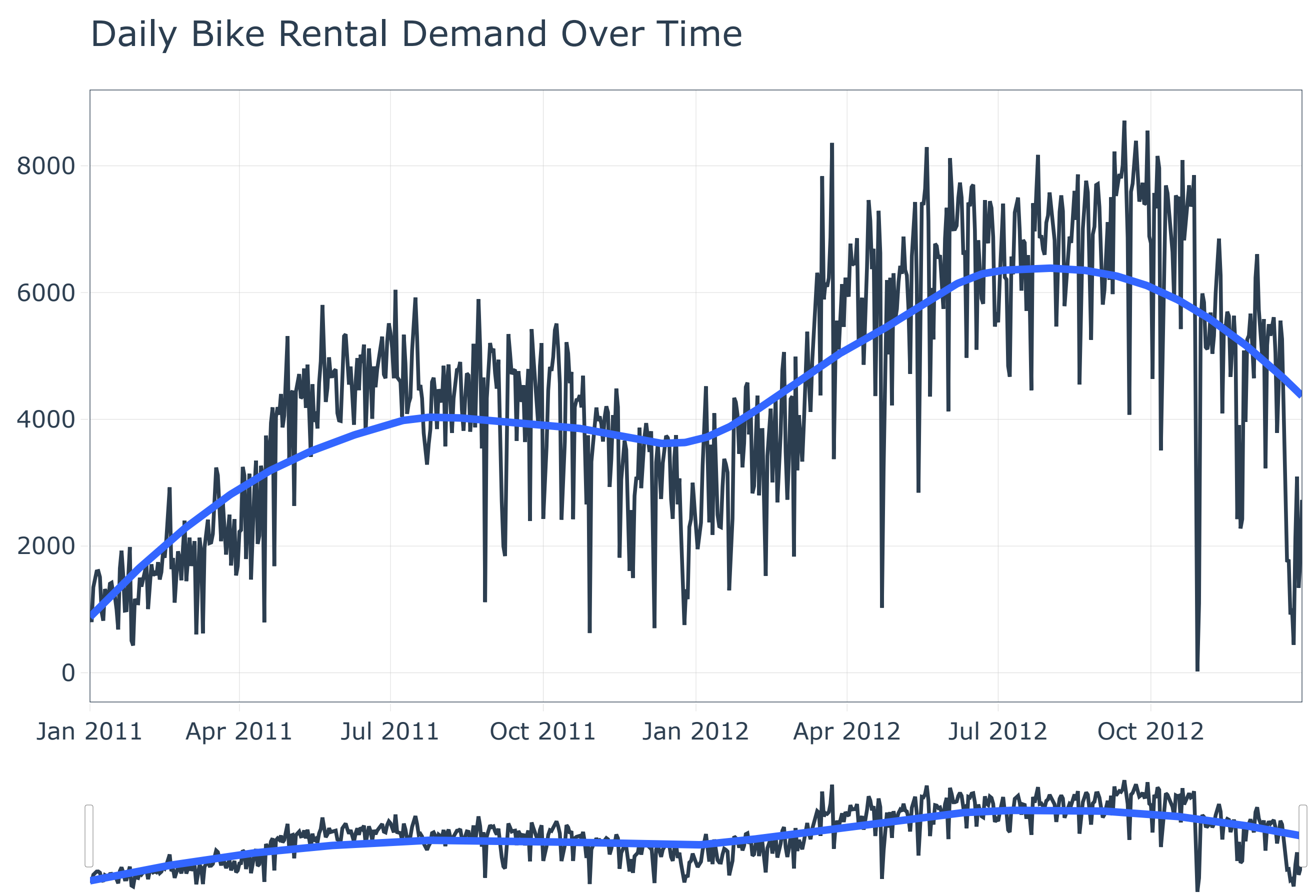
```
data("bike_sharing_daily")
bike_data <- bike_sharing_daily
```

```
bike_data %>%
  mutate(month = lubridate::month(dteday, label = TRUE)) %>%
  group_by(month) %>%
  summarise(
    mean_temp = mean(temp),
    mean_humidity = mean(hum),
    mean_windspeed = mean(windspeed),
    mean_rentals = mean(cnt)
  )
```

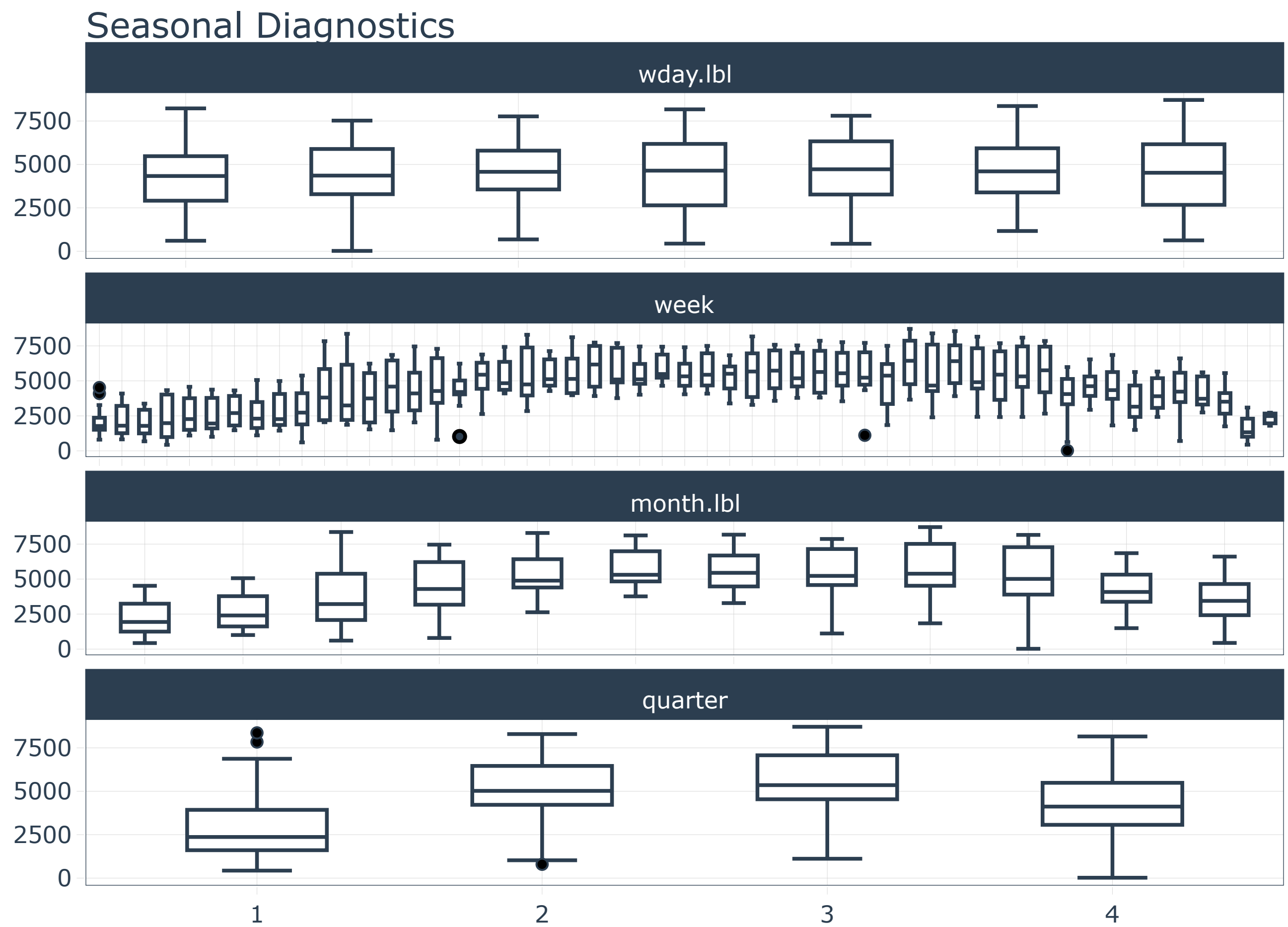
```
## # A tibble: 12 × 5
##    month mean_temp mean_humidity mean_windspeed mean_rentals
##    <ord>     <dbl>         <dbl>          <dbl>        <dbl>
##  1 Jan       0.236         0.586          0.206        2176.
##  2 Feb       0.299         0.567          0.216        2655.
##  3 Mar       0.391         0.588          0.223        3692.
##  4 Apr       0.470         0.588          0.234        4485.
##  5 May       0.595         0.689          0.183        5350.
##  6 Jun       0.684         0.576          0.185        5772.
##  7 Jul       0.755         0.598          0.166        5564.
##  8 Aug       0.709         0.638          0.173        5664.
##  9 Sep       0.616         0.715          0.166        5767.
## 10 Oct       0.485         0.694          0.175        5199.
## 11 Nov       0.369         0.625          0.184        4247.
## 12 Dec       0.324         0.666          0.177        3404.
```

#Task two: Create Interactive Time Seies Plot

```
bike_data %>%
  plot_time_series(
    dteday,
    cnt,
    .interactive = TRUE,
    .plotly_slider = TRUE,
    .title = "Daily Bike Rental Demand Over Time"
  )
```



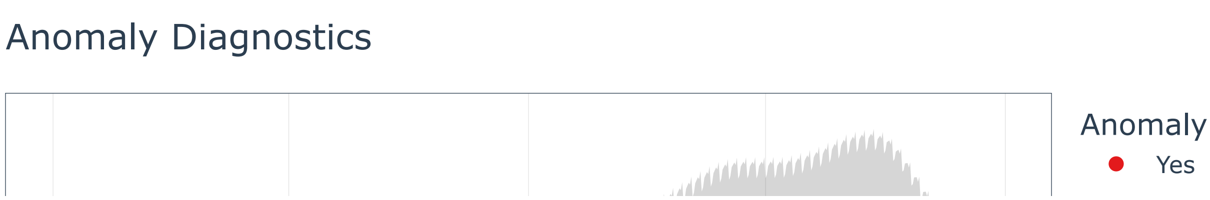
```
bike_data %>%
  plot_seasonal_diagnostics(
    dteday,
    cnt
  )
```

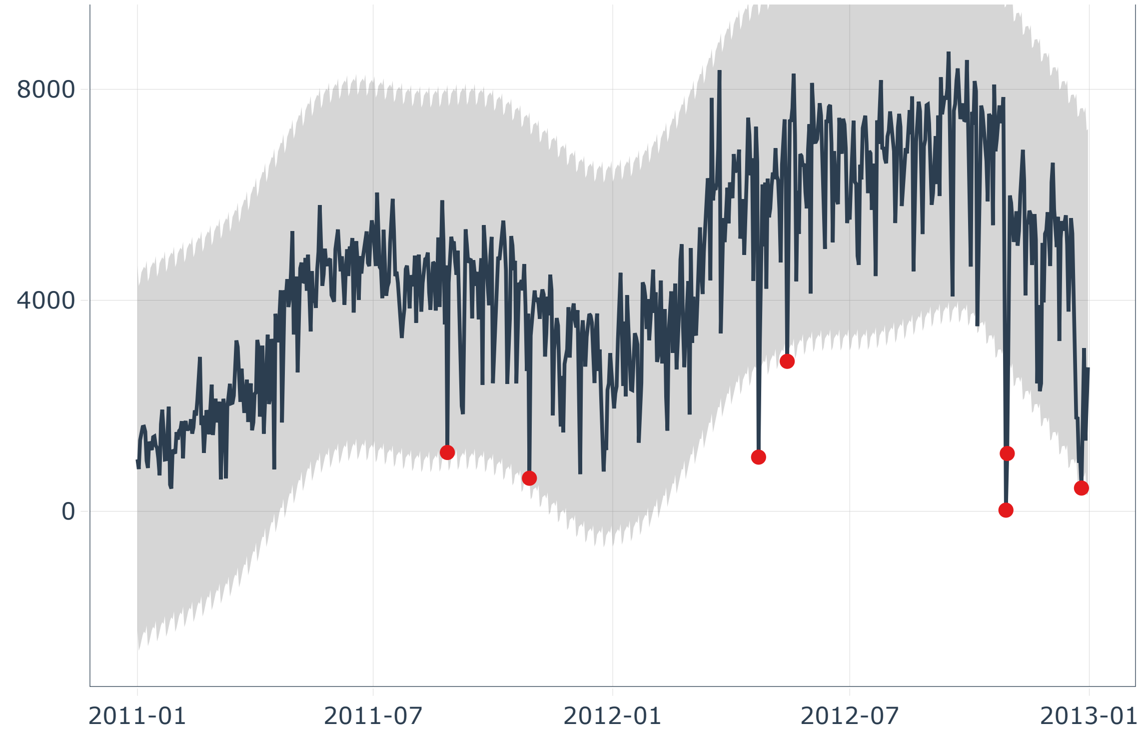



```
bike_data %>%
  plot_anomaly_diagnostics(
    dteday,
    cnt
  )
```

```
## frequency = 7 observations per 1 week
```

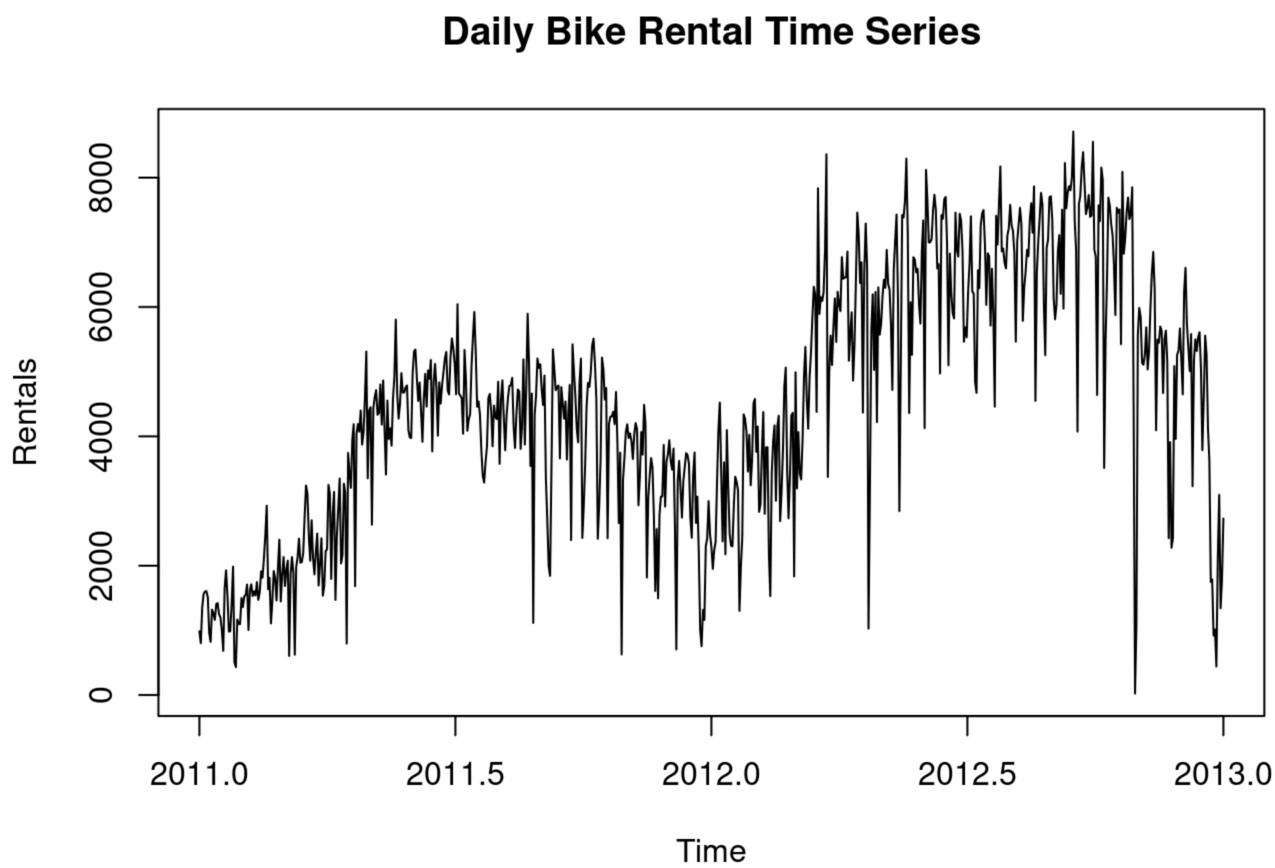
```
## trend = 92 observations per 3 months
```



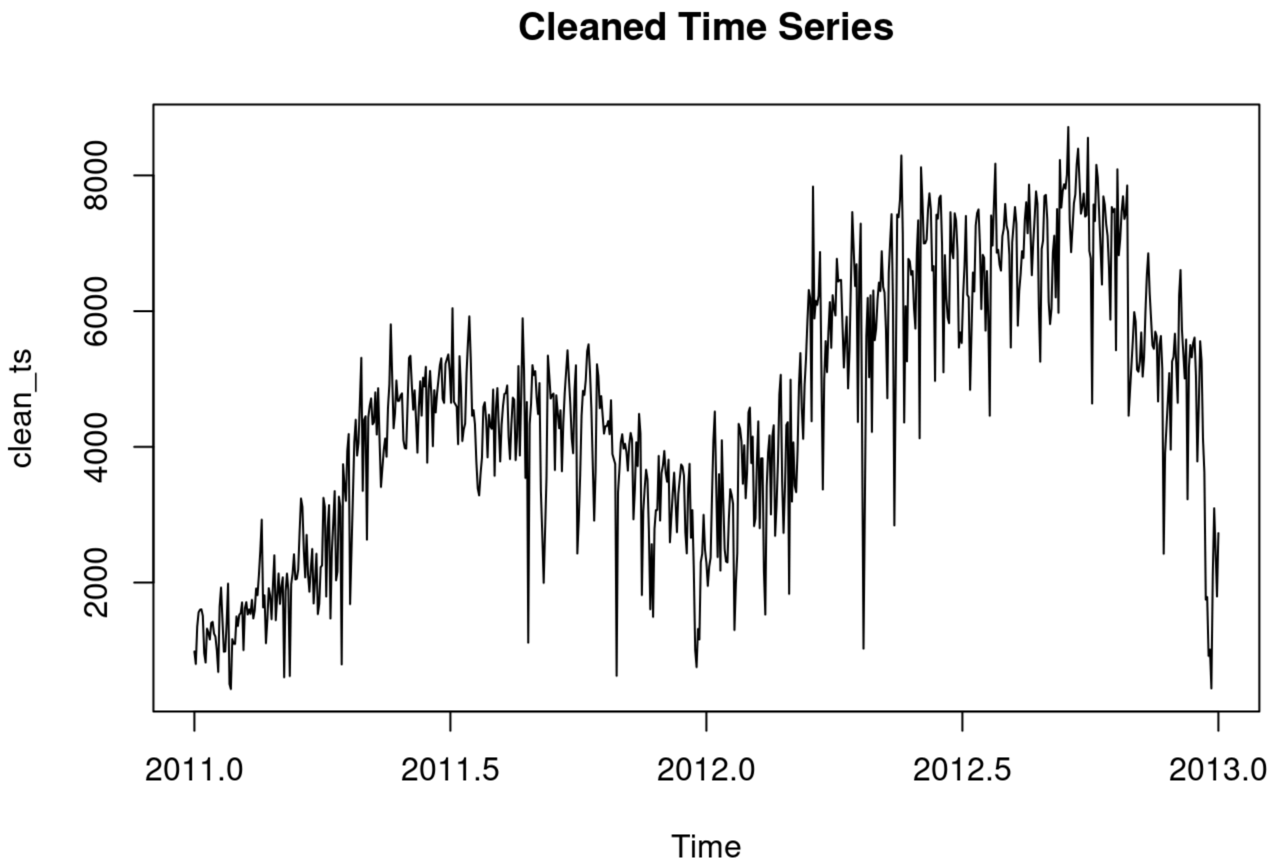


#Task Three: Smooth Time Series Data

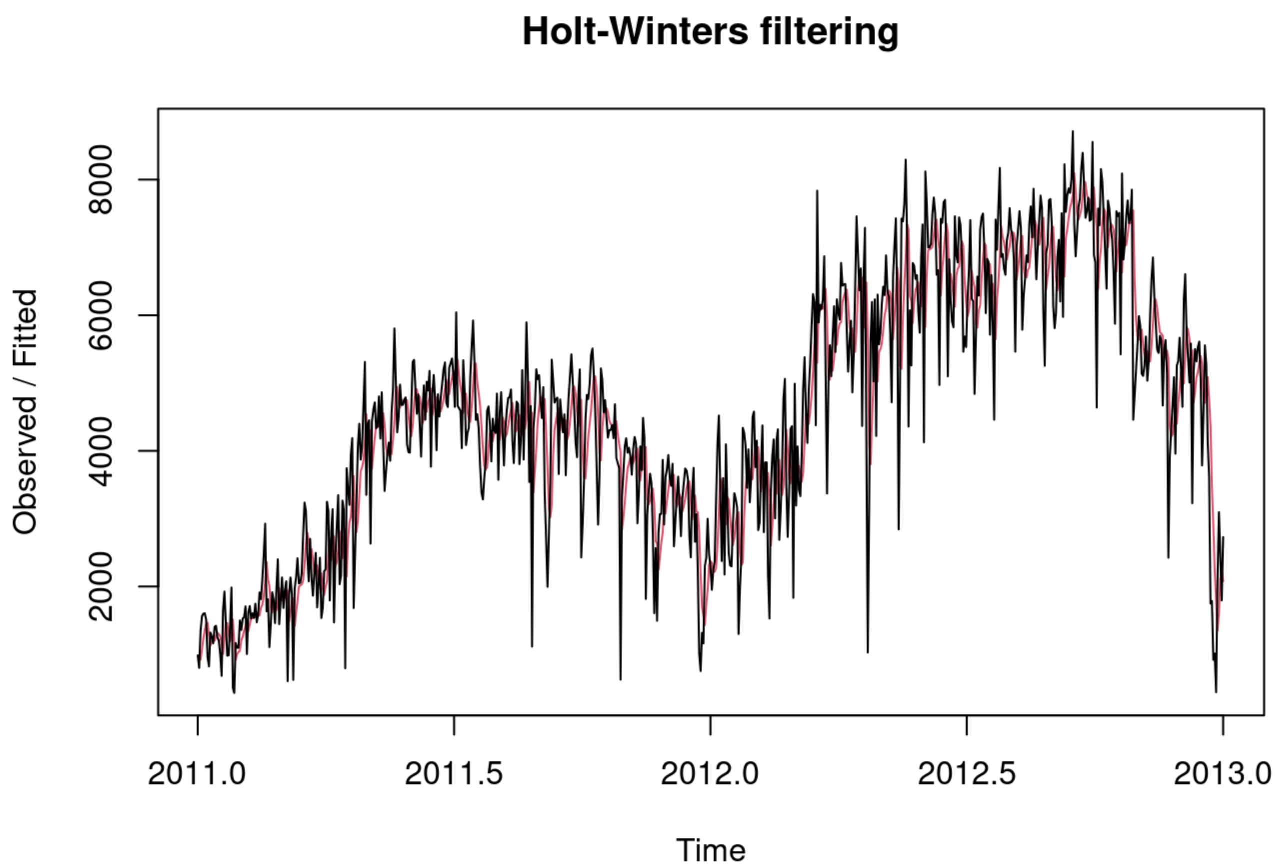
```
ts_data <- ts(bike_data$cnt,
              frequency = 365,
              start = c(2011,1))

plot(ts_data,
     main="Daily Bike Rental Time Series",
     ylab="Rentals",
     xlab="Time")
```



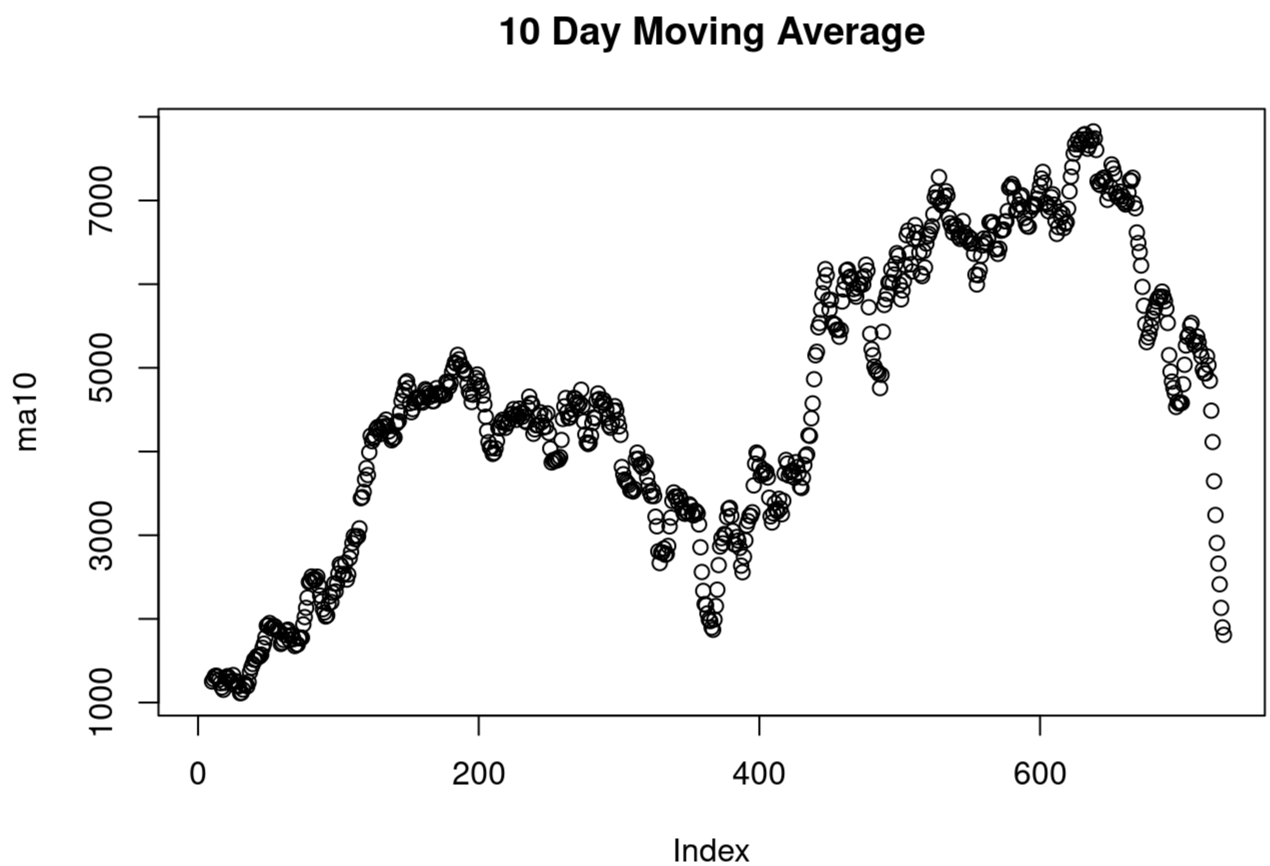

```
clean_ts <- tsclean(ts_data)

plot(clean_ts,
     main="Cleaned Time Series")
```




```
ses_model <- HoltWinters(clean_ts, beta = FALSE, gamma = FALSE)

plot(ses_model)
```

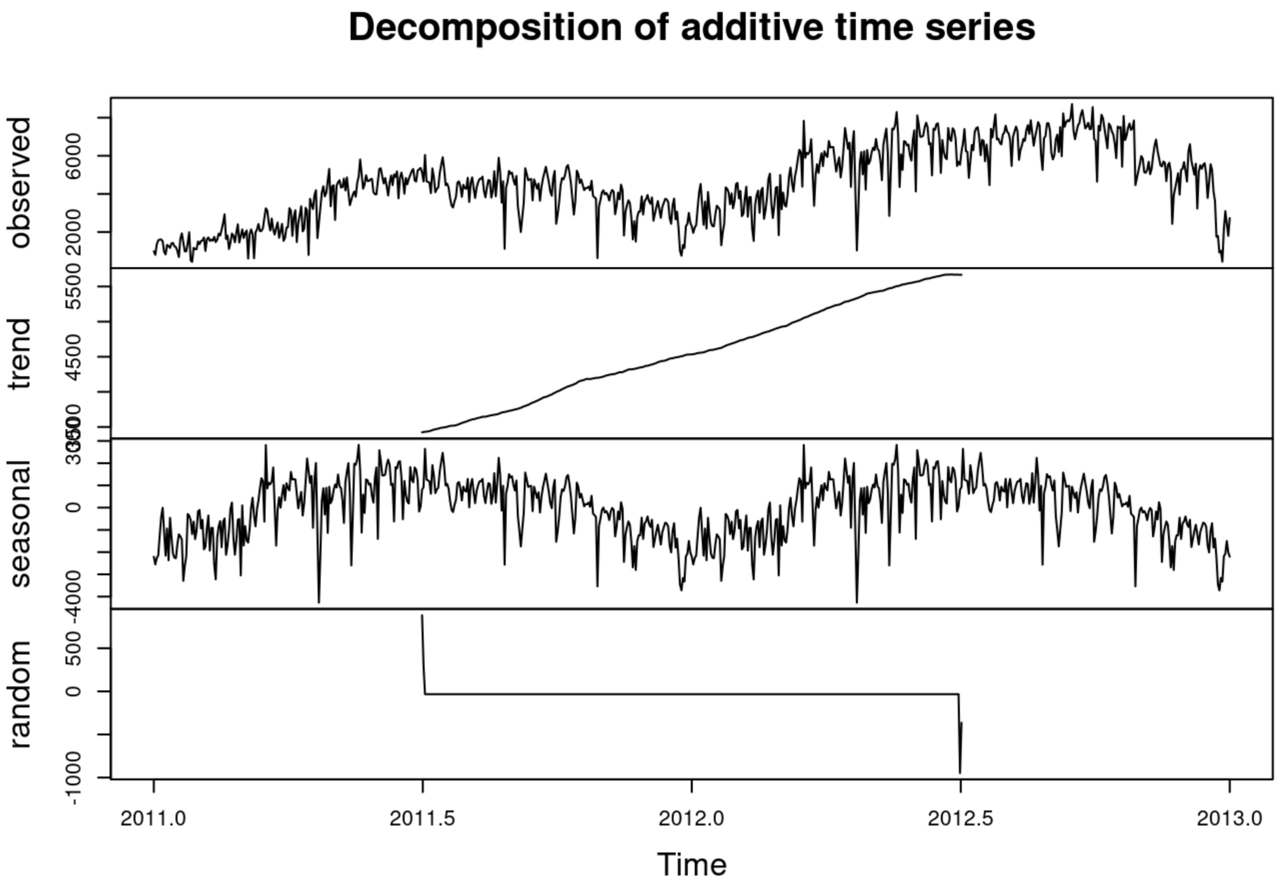



```
ma10 <- SMA(clean_ts, n=10)

plot(ma10,
     main="10 Day Moving Average")
```



#Task Four…

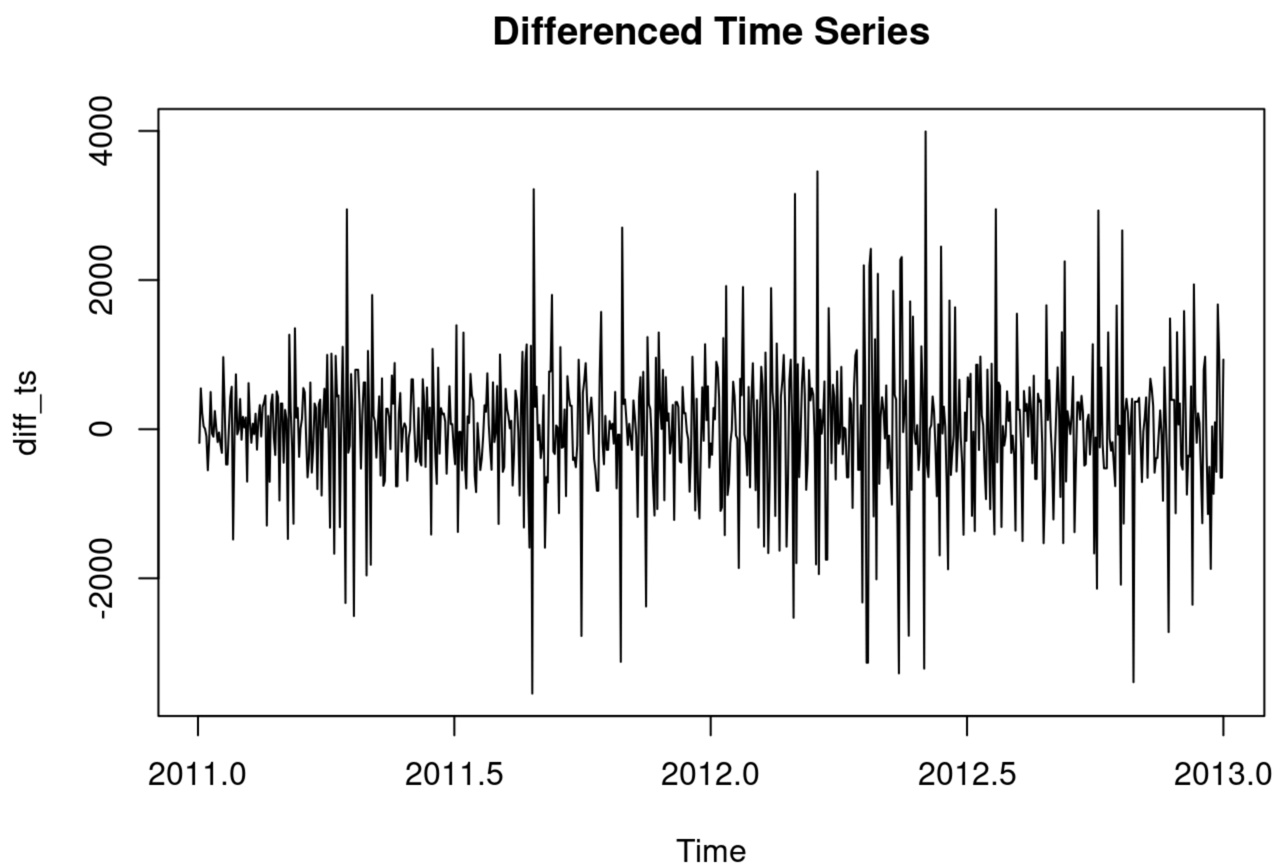
```
decomp <- decompose(clean_ts)

plot(decomp)
```



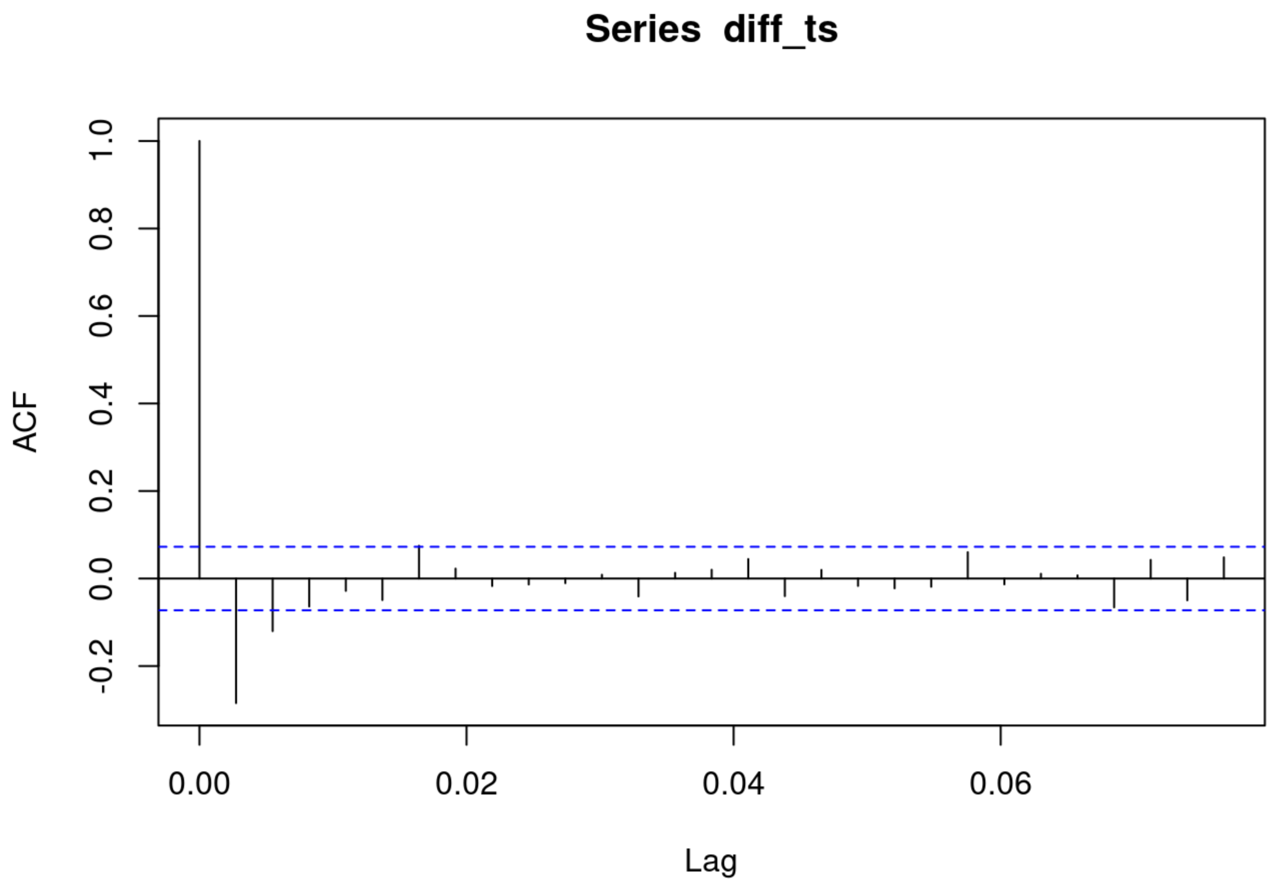

```
adf.test(clean_ts)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  clean_ts
## Dickey-Fuller = -1.4435, Lag order = 9, p-value = 0.8138
## alternative hypothesis: stationary
```

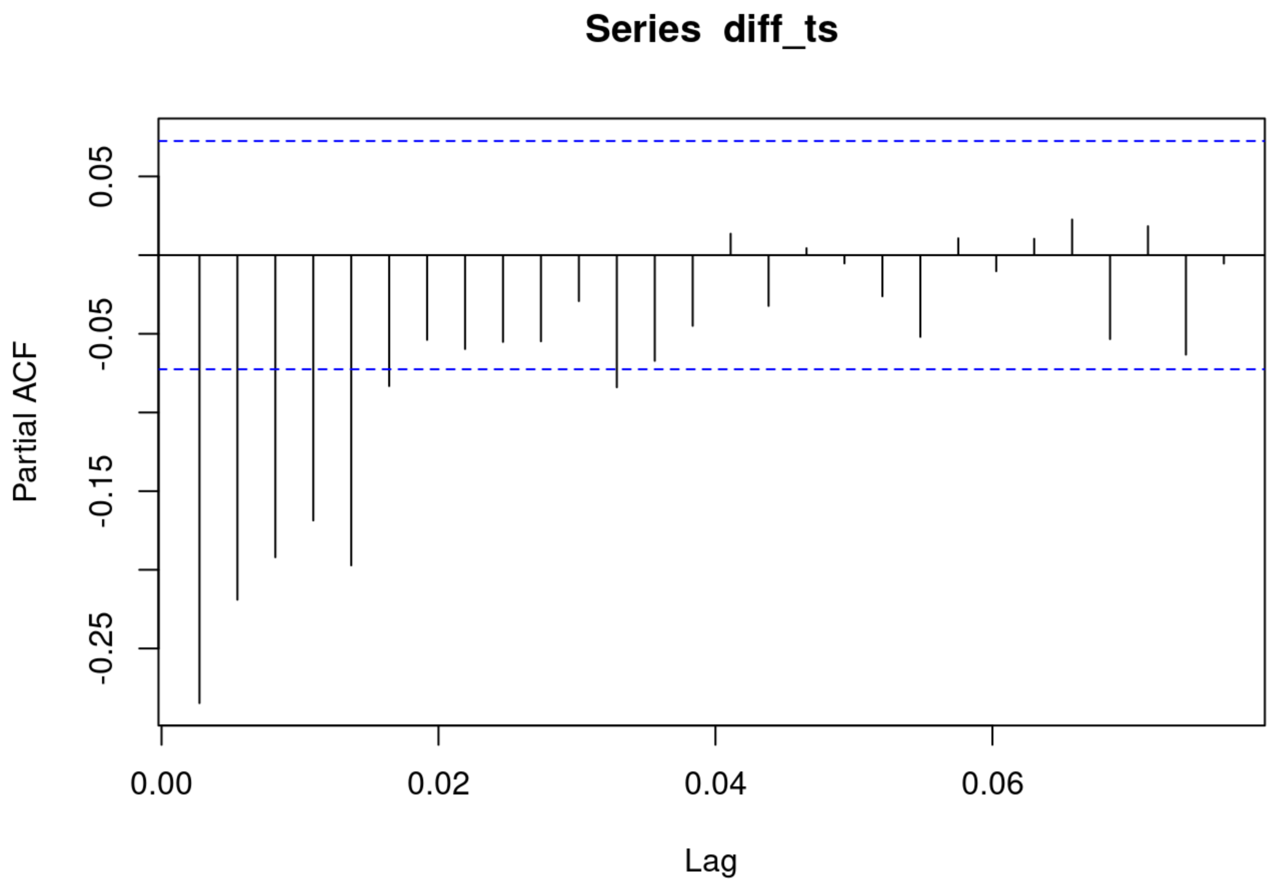
```
diff_ts <- diff(clean_ts)

plot(diff_ts,
     main="Differenced Time Series")
```


Differenced Time Series
diff_ts
4000
2000
0
-2000
2011.0
2011.5
2012.0
2012.5
2013.0
Time

```
acf(diff_ts)
```




```
pacf(diff_ts)
```



#Task Five…

```
clean_ts
```

```
## Time Series:
## Start = c(2011, 1)
## End = c(2013, 1)
## Frequency = 365
##   [1]  985.000  801.000 1349.000 1562.000 1600.000 1606.000 1510.000  959.000
##   [9]  822.000 1321.000 1263.000 1162.000 1406.000 1421.000 1248.000 1204.000
##  [17] 1000.000  683.000 1650.000 1927.000 1454.000  981.000  986.000 1416.000
##  [25] 1985.000  506.000  431.000 1167.000 1098.000 1096.000 1501.000 1360.000
##  [33] 1526.000 1550.000 1708.000 1005.000 1623.000 1712.000 1530.000 1605.000
##  [41] 1538.000 1746.000 1472.000 1589.000 1913.000 1815.000 2115.000 2475.000
##  [49] 2927.000 1635.000 1812.000 1107.000 1450.000 1917.000 1807.000 1461.000
##  [57] 1969.000 2402.000 1446.000 1790.000 2134.000 1685.000 1944.000 2077.000
##  [65]  605.000 1872.000 2133.000 1891.000  623.000 1977.000 2132.000 2417.000
##  [73] 2046.000 2056.000 2192.000 2744.000 3239.000 3117.000 2471.000 2077.000
##  [81] 2703.000 2121.000 1865.000 2210.000 2496.000 1693.000 2028.000 2425.000
##  [89] 1536.000 1685.000 2227.000 2252.000 3249.000 3115.000 1795.000 2808.000
##  [97] 3141.000 1471.000 2455.000 2895.000 3348.000 2034.000 2162.000 3267.000
## [105] 3126.000  795.000 3744.000 3429.000 3204.000 3944.000 4189.000 1683.000
## [113] 2479.667 3276.333 4073.000 4400.000 3872.000 4058.000 4685.000 5312.000
## [121] 3351.000 4401.000 4451.000 2633.000 4433.000 4608.000 4714.000 4333.000
## [129] 4362.000 4803.000 4182.000 4864.000 4105.000 3409.000 3683.500 3958.000
## [137] 4123.000 3855.000 4575.000 4917.000 5805.000 5039.500 4274.000 4492.000
## [145] 4978.000 4677.000 4679.000 4758.000 4788.000 4098.000 3982.000 3974.000
## [153] 4643.000 5312.000 5342.000 4906.000 4548.000 4833.000 4401.000 3915.000
## [161] 4586.000 4966.000 4460.000 5020.000 4891.000 5180.000 3767.000 4844.000
## [169] 5119.000 4744.000 4010.000 4835.000 4507.000 4790.000 4991.000 5202.000
## [177] 5305.000 4708.000 4648.000 5225.000 5293.500 5362.000 5119.000 4649.000
## [185] 6043.000 4665.000 4629.000 4592.000 4040.000 5336.000 4881.000 4086.000
## [193] 4258.000 4342.000 5084.000 5538.000 5923.000 5302.000 4458.000 4541.000
## [201] 4332.000 3784.000 3387.000 3285.000 3606.000 3840.000 4590.000 4656.000
## [209] 4390.000 3846.000 4475.000 4302.000 4266.000 4845.000 3574.000 4576.000
## [217] 4866.000 4294.000 3785.000 4326.000 4602.000 4780.000 4792.000 4905.000
## [225] 4150.000 3820.000 4338.000 4725.000 4694.000 3805.000 4153.000 5191.000
## [233] 3873.000 4758.000 5895.000 5130.000 3542.000 4661.000 1115.000 4334.000
## [241] 4634.000 5204.000 5058.000 5115.000 4727.000 4484.000 4940.000 3351.000
## [249] 2710.000 1996.000 2770.000 3544.000 5345.000 5046.000 4713.000 4763.000
## [257] 4785.000 3659.000 4760.000 4511.000 4274.000 4539.000 3641.000 4352.000
## [265] 4795.000 5109.000 5423.000 5010.000 4630.000 4120.000 3907.000 4839.000
## [273] 5202.000 2429.000 2918.000 3570.000 4456.000 4826.000 4765.000 4985.000
## [281] 5409.000 5511.000 5117.000 4563.000 3738.000 2913.000 3644.000 5217.000
## [289] 5041.000 4570.000 4748.000 4471.500 4195.000 4304.000 4308.000 4381.000
## [297] 4187.000 4687.000 3894.000 3820.500 3747.000  627.000 3331.000 3669.000
## [305] 4068.000 4186.000 3974.000 4046.000 3926.000 3649.000 4035.000 4205.000
## [313] 4109.000 2933.000 3368.000 4067.000 3717.000 4486.000 4195.000 1817.000
## [321] 3053.000 3392.000 3663.000 3520.000 2765.000 1607.000 2566.000 1495.000
## [329] 2792.000 3068.000 3071.000 3867.000 2914.000 3613.000 3727.000 3940.000
## [337] 3614.000 3485.000 3811.000 2594.000 2958.000 3322.000 3620.000 3190.000
## [345] 2743.000 3310.000 3523.000 3740.000 3709.000 3577.000 2739.000 2431.000
## [353] 3403.000 3750.000 2660.000 3068.000 2209.000 1011.000  754.000 1317.000
## [361] 1162.000 2302.000 2423.000 2999.000 2485.000 2294.000 1951.000 2236.000
## [369] 2368.000 3272.000 4098.000 4521.000 3425.000 2376.000 3598.000 2177.000
## [377] 4097.000 3214.000 2493.000 2311.000 2298.000 2935.000 3376.000 3292.000
```

```
## [385] 3163.000 1301.000 1977.000 2432.000 4339.000 4270.000 4075.000 3456.000
## [393] 4023.000 3243.000 3624.000 4509.000 4579.000 3761.000 4151.000 2832.000
## [401] 2947.000 3784.000 4375.000 2802.000 3830.000 3831.000 2169.000 1529.000
## [409] 3422.000 3922.000 4169.000 3005.000 4154.000 4318.000 2689.000 3129.000
## [417] 3777.000 4773.000 5062.000 3487.000 2732.000 3389.000 4322.000 4363.000
## [425] 1834.000 4990.000 3194.000 4066.000 3423.000 3333.000 3956.000 4916.000
## [433] 5382.000 4569.000 4118.000 4911.000 5298.000 5847.000 6312.000 6192.000
## [441] 4378.000 7836.000 5892.000 6153.000 6093.000 6230.000 6871.000 5121.500
## [449] 3372.000 4996.000 5558.000 5102.000 5698.000 6133.000 5459.000 6235.000
## [457] 6041.000 5936.000 6772.000 6436.000 6457.000 6460.000 5814.500 5169.000
## [465] 5585.000 5918.000 4862.000 5409.000 6398.000 7460.000 6915.000 6370.000
## [473] 6691.000 4367.000 6565.000 7290.000 4158.500 1027.000 3214.000 5633.000
## [481] 6196.000 5026.000 6233.000 4220.000 6304.000 5572.000 5740.000 6169.000
## [489] 6421.000 6296.000 6883.000 6359.000 6273.000 5728.000 4717.000 6572.000
## [497] 7030.000 7429.000 6118.000 2843.000 5115.000 7424.000 7384.000 7639.000
## [505] 8294.000 7129.000 4359.000 6073.000 5260.000 6770.000 6734.000 6536.000
## [513] 6591.000 6043.000 5743.000 6855.000 7338.000 4127.000 8120.000 7641.000
## [521] 6998.000 7001.000 7055.000 7494.000 7736.000 7498.000 6598.000 6664.000
## [529] 4972.000 7421.000 7363.000 7665.000 7702.000 6978.000 5099.000 6825.000
## [537] 6211.000 5905.000 5823.000 7458.000 6891.000 6779.000 7442.000 7335.000
## [545] 6879.000 5463.000 5687.000 5531.000 6227.000 6660.000 7403.000 6241.000
## [553] 6207.000 4840.000 5704.500 6569.000 6290.000 7264.000 7446.000 7499.000
## [561] 6969.000 6031.000 6830.000 6786.000 5713.000 6591.000 5870.000 4459.000
## [569] 7410.000 6966.000 7592.000 8173.000 6861.000 6904.000 6685.000 6597.000
## [577] 7105.000 7216.000 7580.000 7261.000 7175.000 6824.000 5464.000 7013.000
## [585] 7273.000 7534.000 7286.000 5786.000 6299.000 6544.000 6883.000 6784.000
## [593] 7347.000 7605.000 7148.000 7865.000 7197.500 6530.000 7006.000 7375.000
## [601] 7765.000 7582.000 6053.000 5255.000 6917.000 7040.000 7697.000 7713.000
## [609] 7350.000 6140.000 5810.000 6034.000 6864.000 7112.000 6203.000 7504.000
## [617] 5976.000 8227.000 7525.000 7767.000 7870.000 7804.000 8009.000 8714.000
## [625] 7333.000 6869.000 7230.000 7591.000 7720.000 8167.000 8395.000 7907.000
## [633] 7436.000 7538.000 7733.000 7393.000 7415.000 8555.000 6889.000 6778.000
## [641] 4639.000 7572.000 7328.000 8156.000 7965.000 7440.667 6916.333 6392.000
## [649] 7691.000 7570.000 7282.000 7109.000 6639.000 5875.000 7534.000 7461.000
## [657] 7509.000 5424.000 8090.000 6824.000 7058.000 7466.000 7693.000 7359.000
## [665] 7444.000 7852.000 4459.000 4828.000 5197.000 5566.000 5986.000 5847.000
## [673] 5138.000 5107.000 5259.000 5686.000 5035.000 5315.000 5992.000 6536.000
## [681] 6852.000 6269.000 5882.000 5495.000 5445.000 5698.000 5629.000 4669.000
## [689] 5499.000 5634.000 5146.000 2425.000 3910.000 4302.333 4694.667 5087.000
## [697] 3959.000 5260.000 5323.000 5668.000 5191.000 4649.000 6234.000 6606.000
## [705] 5729.000 5375.000 5008.000 5582.000 3228.000 5170.000 5501.000 5319.000
## [713] 5532.000 5611.000 5047.000 3786.000 4585.000 5557.000 5267.000 4128.000
## [721] 3623.000 1749.000 1787.000  920.000 1013.000  441.000 2114.000 3095.000
## [729] 2445.500 1796.000 2729.000
```

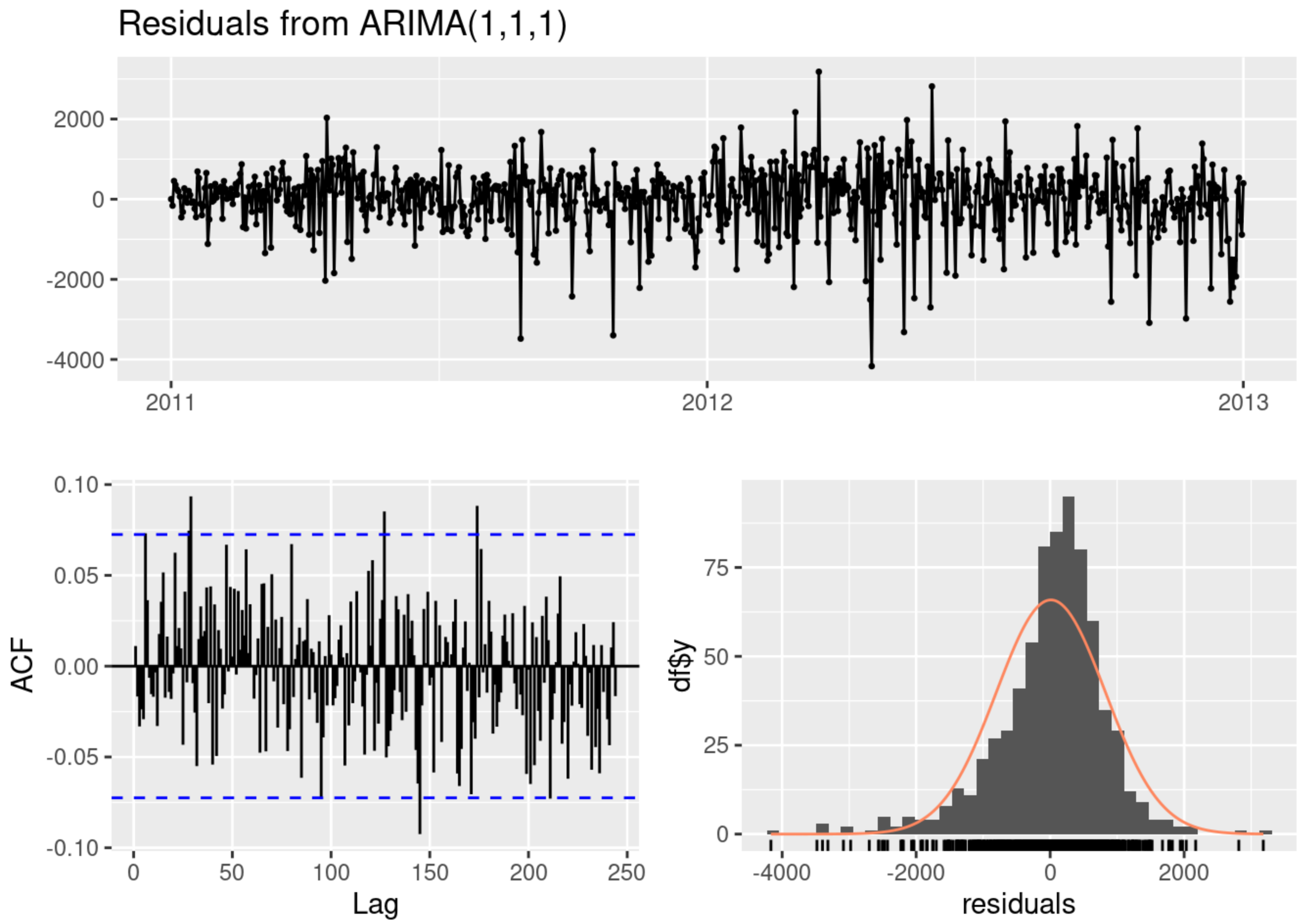
```
auto_model <- auto.arima(clean_ts, seasonal = FALSE)

summary(auto_model)
```

```
## Series: clean_ts
## ARIMA(1,1,1)
##
## Coefficients:
##          ar1      ma1
##       0.3692  -0.8751
## s.e.  0.0437   0.0213
##
## sigma^2 = 663543:  log likelihood = -5928.18
## AIC=11862.37   AICc=11862.4   BIC=11876.15
##
## Training set error measures:
##                    ME     RMSE      MAE       MPE     MAPE      MASE       ACF1
## Training set 11.13247 812.9082 588.1202 -6.429274 19.42943 0.2582884 0.01109983
```

```
manual_model <- arima(clean_ts, order=c(2,1,2))

summary(manual_model)
```

```
##
## Call:
## arima(x = clean_ts, order = c(2, 1, 2))
##
## Coefficients:
##          ar1      ar2      ma1     ma2
##       0.8975  -0.2461  -1.3898  0.4637
## s.e.  0.4065   0.1443   0.4101  0.3562
##
## sigma^2 estimated as 659903:  log likelihood = -5927.19,  aic = 11864.38
##
## Training set error measures:
##                    ME     RMSE     MAE       MPE    MAPE     MASE         ACF1
## Training set 9.057166 811.7884 583.976 -6.507784 19.3814 0.887764 -0.004821705
```

```
checkresiduals(auto_model)
```

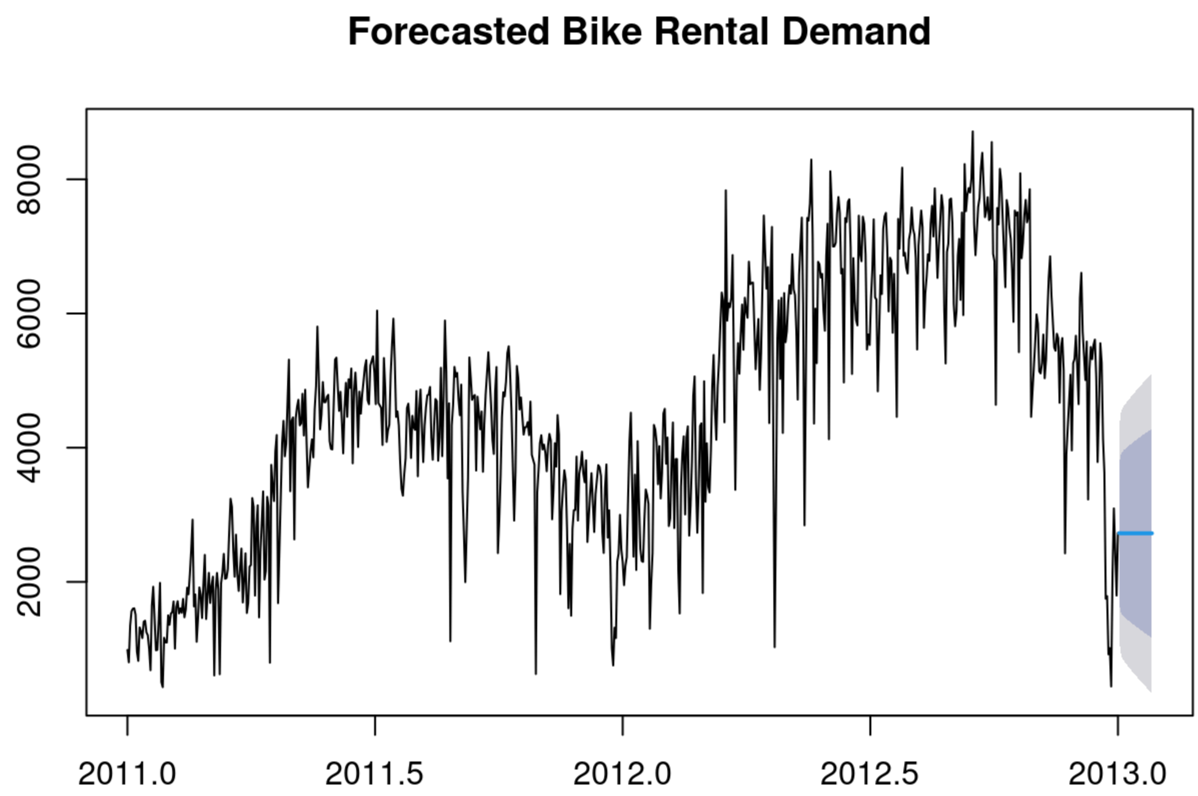



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,1,1)
## Q* = 142.67, df = 144, p-value = 0.5156
##
## Model df: 2.   Total lags used: 146
```

```
shapiro.test(residuals(auto_model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(auto_model)
## W = 0.94187, p-value = 2.596e-16
```

```
forecast_auto <- forecast(auto_model, h=25)

plot(forecast_auto,
     main="Forecasted Bike Rental Demand")
```



Task Six: Findings and Conclusions

Key Insights

- Bike rental demand shows strong seasonal behavior throughout the year.
- Temperature has a positive relationship with bike rental demand. Higher temperatures tend to increase bike usage.
- Rental demand is highest during warmer seasons such as summer, while demand decreases during colder months.
- Registered users contribute significantly more rentals compared to casual users, indicating consistent usage by regular commuters.

Forecast Insights

The ARIMA model successfully captured the underlying trend and seasonal patterns in the bike rental dataset. The forecast results suggest that bike rental demand will continue to follow seasonal trends with fluctuations influenced by weather conditions and seasonal changes.

Business Recommendations

Based on the results of this analysis, the bike rental company can consider the following strategies:

- Increase bike availability during warmer months when rental demand is high.
- Reduce the number of bikes available during colder seasons to optimize operational costs.
- Use demand forecasting to implement dynamic pricing strategies during peak demand periods.
- Expand bike stations in high demand areas to improve service availability and customer satisfaction.

Limitations

- The dataset only contains two years of data, which may limit the long-term forecasting accuracy.
- External factors such as holidays, special events, and economic conditions were not included in this analysis.
- Weather prediction uncertainty may affect the accuracy of future demand forecasts.

Future Work

Future improvements to this analysis could include:

- Incorporating additional weather variables such as precipitation and visibility.
- Applying machine learning forecasting models such as Random Forest or Gradient Boosting.
- Analyzing hourly bike rental demand to obtain more detailed insights into user behavior.

Conclusion

Time series analysis provides valuable insights into bike rental demand patterns. By applying ARIMA forecasting models, bike rental companies can improve operational planning, optimize fleet allocation, and enhance pricing strategies. These insights can help businesses better manage resources and improve customer service.